

Homework 1 – Marine Geology

Total 10 points

Due date: Tuesday 07/05/2022 at 2 pm (class time)

- 1- Explain why stars are important to the ocean's formation, composition, and characteristics (2 Pts).**
- 2- Read Elkins-Tanton (2013) article and answer: If both Earth and Venus were probably built from similar rocky materials and formed by similar mass-accretion processes, why is it that Earth has a marine oceanic ecosystem and viable atmosphere whereas Venus is lacking both features? (2 Pts)**
- 3- Describe the theory of plate tectonics and how plate movement has reshaped the planet's surface over time. (2 pts)**
- 4- Describe the various zones that the seabed is generally divided into and how scientists study the ocean floor. (2 pts)**
- 5- Based on the discussion of Heffernan (2019) (2 pts):**
 - a. Even though seabed mining may be economically lucrative, what are the biological concerns related to such largescale disturbance of the seafloor?**
 - b. What type of sediments are the deep-sea miners interested in and why?**
 - c. Are mid-ocean ridges the only deep-ocean feature where sulfite deposits form? Based on the Plate Tectonics class, where can we expect such minerals to form?**
 - d. Can we expect to find the valuable minerals over the continental shelf? Why?**